

Journal Pre-proofs

Thermodynamic simulation on indium enrichment condition in the Huangshaping skarn Zn-Pb system (South China)

Yihan Wu, Yi Zheng, Pengpeng Yu, Yin Huang, Jie Yu, Taiping Zhao, Qiuming Cheng, Xi Chen

PII: S1367-9120(25)00196-8
DOI: <https://doi.org/10.1016/j.jseaes.2025.106681>
Reference: JAES 106681

To appear in: *Journal of Asian Earth Sciences*

Received Date: 25 January 2025
Revised Date: 8 April 2025
Accepted Date: 29 May 2025

Please cite this article as: Wu, Y., Zheng, Y., Yu, P., Huang, Y., Yu, J., Zhao, T., Cheng, Q., Chen, X., Thermodynamic simulation on indium enrichment condition in the Huangshaping skarn Zn-Pb system (South China), *Journal of Asian Earth Sciences* (2025), doi: <https://doi.org/10.1016/j.jseaes.2025.106681>

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2025 Published by Elsevier Ltd.



**Thermodynamic simulation on indium enrichment condition in the
Huangshaping skarn Zn-Pb system (South China)**

Yihan Wu¹, Yi Zheng^{1, 2, 3}, Pengpeng Yu^{1, 2}, Yin Huang¹, Jie Yu⁴, Taiping Zhao⁵, Qiuming Cheng¹, Xi Chen^{1*}

¹ *Guangdong Provincial Key Lab of Geological Process and Mineral Resources Survey, School of Earth Sciences and Geological Engineering, Sun Yat-sen University, Guangzhou 510275, China.*

² *Guangdong Provincial Key Lab of Geodynamic and Geohazards, Sun Yat-sen University, Guangzhou 510275, China.*

³ *Southern Laboratory of Ocean Science and Engineering, Zhuhai 519000, China*

⁴ *School of Earth and Planetary Sciences, Curtin University, Perth, WA, 6845, Australia.*

⁵ *Key Laboratory of Mineralogy and Metallogeny, Guangzhou Institute of Geochemistry, Chinese Academy of Sciences, Guangzhou 510640, China.*

***Corresponding author:** Xi Chen

Address: Building of Geology and Environment, Sun Yat-sen University, Guangzhou 510275, China. Tel.: +86 15201660942.

Email: chenx783@mail.sysu.edu.cn (*X. Chen*)

Abstract

Indium, crucial for emerging industries like optoelectronics, is a strategic metal linked to national prosperity and defense. However, understanding its enrichment during migration-precipitation in skarn-type Zn-Pb deposits is limited. This study focuses on the Huangshaping Zn-Pb deposit in the Nanling belt, China, which is a skarn-type deposit with quartz porphyry as the ore-causative intrusion. Through petrography, mineralization was identified as skarn, oxide, and sulfide stages, with three sphalerite types (Sp1, Sp2a, Sp2b). EPMA and LA-ICP-MS analyses revealed different indium concentrations, with Sp1 being the most enriched (mean = 134.54 ppm). LA-ICP-MS mapping showed that Sp1 is rich in In, Cu, and Sn. Mineral paragenesis combined with GGIMFis thermometer for sphalerite trace elements and thermodynamic calculations indicate indium-rich sphalerite precipitates under weakly acidic pH 4.5–6.0, high oxygen fugacity ($\log f_{O_2} = -32.2$ to -29.7), high temperature (201–316°C), and high sulfur fugacity ($\log f_{S_2} = -13.42$ to -7.11) in a reduced, H₂S-rich environment. We propose that in skarn systems, the enrichment of indium occurs during the early stages of sphalerite crystallization, specifically at high temperatures and in deeper spatial positions. This enrichment process occurs in a fluid environment characterized by a weakly acidic pH, high oxygen fugacity, and high sulfur fugacity. The findings deepen our understanding on indium enrichment and the physicochemical conditions

in skarn-type Zn-Pb systems and may aid prospecting in similar deposits.

Keywords: Indium; Sphalerite; Skarn; Thermodynamic simulation.

1 Introduction

Indium (In) is a crucial metal that has been widely and substantially applied in many advanced technology fields, including liquid crystal displays (LCDs), touch screens featuring indium tin oxide (ITO) coatings, solar panels, and the nuclear field (Werner et al., 2017). From a geochemical perspective, indium is characterized as a dispersed element with a relatively low abundance (0.056 ppm in the upper crust, Rudnick and Gao, 2003).

The In is regarded as a strongly chalcophile element (Greaney et al., 2017). Moreover, it exhibits moderately incompatible behavior during partial melting (Yi et al., 2000), preferentially enriching in base metal sulfides. The demand for indium is expected to increase over the next few decades. Sphalerite provides 95% of the global indium supply (Ishihara et al., 2011; Li et al., 2023; Werner et al., 2017). As an important global indium-producing country, China supplied nearly half of the total global refined indium during the decade from 2011 to 2021 (USGS, 2021). This underscores China's pivotal role in producing and supplying indium resources and highlights the strategic importance of indium resources within the global industrial landscape and economic growth trajectory.

Recent work has suggested that indium-rich Zn-Pb deposits are often

closely related to granite, such as magmatic hydrothermal deposits with relatively high mineralization temperatures (Zhang, 1987, 1998; Cook et al., 2009; Ye et al., 2011; Zhao et al., 2022). In magmatic hydrothermal deposits, indium shows a remarkable feature of being enriched in the early stage and in deep spaces (Wu et al., 2023). Based on the hard and soft acid-base theory (Pearson, 1963), In^{3+} , as a hard acid, can form stable complexes with hard bases such as OH^- , F^- , Cl^- , and SO_4^{2-} and mainly migrates in fluids as chlorine complexes, hydrates and fluorides (Seward et al., 2000; Wood et al., 2006). Meanwhile, the enrichment of indium is more inclined to occur in hydrothermal fluid environments with relatively high temperatures ($\geq 300^\circ\text{C}$) and low salinities (< 10 wt.% NaCl eq.; Kumar et al., 2023). Moreover, carbonate rocks can also change the pH of the fluid to further facilitate the precipitation of indium-rich sulfides (Liu et al., 2023). In addition, the concentrations and distributions of these trace elements in sphalerite are controlled by a series of complex physicochemical parameters, such as the chemical composition of ore fluids, temperature, redox, and pH-Eh conditions (Scott & Barnes, 1971; Gagnevin et al., 2014; Lockington et al., 2014; Frenzel et al., 2016). However, during the process from the migration to the precipitation of ore-forming fluids, how the physicochemical conditions regulate the extraordinary enrichment of indium still needs to be further investigated and analyzed.

The Huangshaping polymetallic deposit is located in the middle of the Nanling W-Sn-Zn-Pb polymetallic metallogenic belt in southern China (Fig. 1). This deposit is a typical one among the eleven giant skarn lead-zinc deposits in the Nanling region. It contains abundant metal resources such as copper, molybdenum, lead and zinc, and is also associated with the economically valuable element indium (Zhang et al., 1998; Ishihara and Murakami, 2008; Ishihara et al., 2011; Wu et al., 2023). Previous studies on the critical metal content of the Huangshaping polymetallic deposits have been done. Pan et al. (2011) compared trace elements in certain rocks in the deposit, finding high Ga in granite porphyry (mean = 26.73 ppm). Zhang (2015) analyzed whole-rock trace elements, showing high In in sulfide ores (mean = 33.2 ppm) and Ga in granite porphyry (mean = 20.1 ppm). Yuan (2015) also noted high Ga in granite porphyry (about 29 ppm). Zhao (2017) found high In in sphalerite (0.18–0.46 wt.%) from the ore zone. Zhu et al. (2021) tested magnetite in sulfides with LA-ICP-MS and measured Ga content at ~ 100 ppm. However, the physicochemical conditions for the migration and precipitation of In-rich ore-forming fluids remain unknown.

This study aims to comprehensively integrate field geological surveys, petrographic analysis, electron probe microanalyzer (EPMA) technology, laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) detection methods and thermodynamic modeling methods in order to

precisely delineate the physicochemical conditions involved in the migration and precipitation of indium-rich ore from mineralizing fluids, providing essential physicochemical constraints necessary to elucidate the metallogenic mechanism of this indium-rich deposit..

2 Regional geology

The South China Block (SCB) is an important base for rare and rare earth mineral resources in China, especially with the extraordinary enrichment of tungsten, tin, lithium, beryllium, niobium, and tantalum (Mao et al., 2013; Liu et al., 2017; Yu et al., 2023). This block is located on the southeastern margin of the Eurasian Plate, adjacent to the Indian Plate in the west and the Pacific Plate in the east (Fig. 1a). It was formed by the Neoproterozoic amalgamation of the Yangtze Craton and Cathaysia Block (Shu and Charvet, 1996; Zhou et al., 2018; Yu et al., 2023). The block has experienced multi-stage tectonic and magmatic events, and its basement is mainly composed of Paleoproterozoic and Meso- to Neoproterozoic metamorphic rocks. It has seen multi-stage tectonic and magmatic events, with a basement of Paleo- and Meso- to Neoproterozoic metamorphic rocks (Zhao and Cawood, 1999; Li et al., 2009; Wang et al., 2014). Many SCB granites result from Mesozoic magmatic activities, mainly in the Triassic, Middle-Late Jurassic, and Cretaceous (Fig. 1b, Zhou et al., 2006). The Jurassic activity is vital for polymetallic formation, with Middle-Late Jurassic granitoids divided into two types related to different

polymetallic mineralization (Hong et al., 2022; Huang et al., 2023; Wang et al., 2023).

The Nanling Metallogenic Belt (NMB) is located in the central area of the SCB and is famous for its world-class W-Sn-Zn-Pb polymetallic mineralization. Its metal reserves play a significant role in China, accounting for approximately 73% of China's tungsten, 85% of tin, 30% of lead, and 22% of zinc reserves (Fig. 1b; Wang et al., 2007). For regional structure, faults in the NE, NW, and EW directions are well developed, corresponding to the Guangji-Pingnan, Jiexi-Xiangxiang, and Xianyou-Zhangping faults, respectively (Fig. 1b). Jurassic granite intrusions exist in various forms such as batholiths, sills, stocks, and dikes, and most of them are medium-grained granites (Shu et al., 2011; Yuan et al., 2011; Zhao et al., 2017). The representative deposits in this region include the Baoshan Cu-Zn-Pb deposit, the Huangshaping Zn-Pb polymetallic deposit, the Xianghualing Sn-Zn-Pb deposit, and the Furong Sn deposit. Studies have shown that the formation of the polymetallic deposits in the Nanling metallogenic belt is closely related to the large-scale Mesozoic magmatic activities under the geodynamic background of the subduction of the Paleo-Pacific Ocean. Metal-rich magmatic hydrothermal fluids constitute the source of various metals (Zhou et al., 2006; Mao et al., 2013). Meanwhile, various deposit types have been successively proposed, such as skarn, epithermal, and intrusion-related hydrothermal deposits (Lu et al., 2017;

Ding et al., 2022; Wu et al., 2023). Nevertheless, specific physicochemical parameters of magmatic hydrothermal deposits, such as temperature, sulfur fugacity, and redox state, still need to be further explored and clarified.

3 Ore deposit geology

The Huangshaping Zn-Pb polymetallic deposit, located in the southern NMB, covers less than 3 square kilometers and is rich in tungsten, molybdenum, and lead-zinc (Ding et al., 2018). It contains 1.5291 Mt of Zn (7.13% grade), 761.3 Kt of Pb (3.55% grade), 152.9 Kt of WO_3 (0.2% grade), and 43.2 Kt of Mo (0.2% grade) (Ding et al., 2016). The host rocks are marine carbonate formations. The Upper Devonian Xikuangshan Formation and the Lower Carboniferous Douling'ao, Shidengzi, Ceshui, and Zimenqiao Formations are exposed (Fig. 2; Zhu et al., 2021; Chu et al., 2022). The Zimenqiao and Shidengzi Formations are the main hosts for Zn-Pb orebodies (Fig. 2; Ding et al., 2016). The ore-controlled structures center around the axial area of the Baoling overturned anticline. There are shearing fractures (NE-NNE) like F1, F2, F3, and F5 (Fig. 2, east-west, dipping east about 50°) and tensional fractures (near EW, such as F7, F6 and F9 dipping east 50° – 60°), which control the strata contact and play a key role in metallogenesis (Fig. 2; Zhu et al., 2021). The magmatic rocks are granitoids (Fig. 2, Fig. 3), including granite porphyry (in the south, with a concealed stock body near the Baoling intrusion) and quartz porphyry (in the north and center, in laccolith form, dumbbell-shaped, 0.25 square

kilometers, funnel-shaped in section, and dipping east about 50°). The alteration features diverse and zoned types. Skarnization occurs at the contact of granite porphyry and carbonate rocks (Fig. 3). Chloritization is in skarn and contact zones (Fig. 3). Calcitization is in skarn and related ore bodies (Fig. 3). The ore bodies are mainly skarn-type and vein-type lead-zinc (+silver) ores (Ding et al., 2018; Zhu et al., 2021). Among them, skarn-type ore bodies are often found in and around the granite porphyry and contain mineralizations such as iron, bismuth, tungsten, molybdenum, and fluorite (Ding et al., 2016). Relatively speaking, lead-zinc ore bodies are mainly hosted in limestone strata (Fig. 3).

4 Sampling and analytical methods

4.1 Sampling

Three exploited ore levels (-176m, -136m, and 92m) and drilling core (120m) of the Huangshaping deposit were investigated. A total of 120 representative samples were collected, including a range of fresh, unaltered to weakly altered granodiorite porphyry and host rocks. They were made into thin sections and ore blocks. From them, we selected 15 sphalerite specimens for further petrography, and their characteristics, mineral assemblages, and ore textures are summarized in Table 1.

4.2 EPMA analysis

Quantitative sphalerite compositional data were obtained through a JEOL JXA-iSP100 electron probe micro-analyzer, furnished with four

wavelength-dispersive spectrometers, at the Laboratory of Marine Element and Isotope Facilities in Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai). For quantitative analyses, the operating conditions were set as an accelerating voltage of 20 kV, a probe current of 20 nA, and a 5 μm beam. The X-ray lines and corresponding standards were: S K α (Sphalerite), Zn K α (Sphalerite), Fe K α (Marcasite), Sn L α (Tin), In L α (Indium Phosphide), Cd L α (Cadmium), Cu K α (Copper), and Mn K α (Manganese). The peak counting time was 10 s for major elements (S, Zn, Fe) and 50 s for minor elements (Sn, In, Cd, Cu, Mn). The background counting time was half of the peak counting time at high- and low-energy background positions. The ZAF correction routines were applied to reduce the analytical results. The minimum detection limits in sphalerite were S (58 ppm), Zn (221 ppm), Fe (155 ppm), Sn (95 ppm), In (89 ppm), Cd (92 ppm), Cu (111 ppm), and Mn (75 ppm).

4.3 In situ LA-ICP-MS trace elements analysis (spot)

Sphalerite trace elements were measured by an NWR 193 nm ArF Excimer laser-ablation system connected to an iCAP RQ (ICPMS) at Guangzhou Tuoyan Analytical Technology Co., Ltd. in Guangzhou, China. Helium carrier gas at a rate of 0.7 L/min was introduced into the cup, and the aerosol was then mixed with 0.89 L/min argon make-up gas. The laser fluence was 3.5 J/cm², with a repetition rate of 6 Hz and a spot size of 30 μm . Each spot was analyzed for 45s, followed by a 40s background

measurement. A total of sixteen elements of interest were analyzed, namely ^{34}S , ^{66}Zn , ^{57}Fe , ^{55}Mn , ^{111}Cd , ^{65}Cu , ^{115}In , ^{118}Sn , ^{121}Sb , ^{208}Pb , ^{71}Ga , ^{107}Ag , ^{75}As , ^{202}Hg , ^{209}Bi , and ^{74}Ge . Two reference materials, NIST SRM610 and MASS-1, were employed to calibrate the sphalerite trace element compositions, and the sulfide standard SRM 612 and GE8 were used as unknown samples to assess the data quality (Wilson et al., 2002). The software IOLITE (Paton et al., 2011) was utilized to process the analytical signals and obtain reliable trace element compositions. The detailed process can be found in Liu et al. (2008) and Chu et al. (2023).

4.4. In-situ LA-ICP-MS trace elements analysis (mapping)

The instrument and analyzed elements for mapping analysis were identical to those used in spot analysis. The area of selected sphalerite grains was covered with a series of closely-spaced raster lines according to the method of Large et al. (2009). The lines were ablated using a beam size of 15 μm and a repetition rate of 10 Hz. The depth of ablation during mapping was approximately 5 μm . Background levels and the primary sulfide standard were measured both before and after each map. The fifteen monitored elements were the same as in the spot analysis. The mapping reflects the distributions and concentrations of the elements, irrespective of whether they are in the form of nano particles, solid solutions, or micro-inclusions (Large et al., 2009).

4.5 Log fO_2 -pH thermodynamic calculation

The CHNOSZ package, an open-source R package widely applied in geochemistry (Dick, 2019), was utilized to calculate the log fO_2 -pH diagrams of candidate minerals. Primarily used for thermodynamic calculations, it has a comprehensive thermodynamic database named OBIGT by default (Dick, 2019). CHNOSZ incorporated the aqueous species and mineral models proposed by Berman and Brown (1985) and integrated the revised aqueous species equations of Helgeson-Kirkham-Flowers (Helgeson et al., 1981). These characteristics enabled in-depth exploration of the physicochemical conditions of indium-rich ore-forming fluids. Based on petrographic results, the mineral assemblages in the early and middle sulfide stages were taken as research objects and set as Fe-As-Cu-Zn-Pb-S-O-H-Cl and Fe-Cu-Zn-Pb-S-O-H-Cl, respectively. For specific parameter settings, the temperature range for calculating trace elements in sphalerite was from 300°C to 200°C, $P = 1000$ bar, $\Sigma S = 0.1$ m, $\Sigma NaCl = 1$ m, $pH = 0-14$, and the log fO_2 range was determined by combining previous research findings (Zhu et al., 2022) and the presence of calcite in fluid inclusions (Huang et al., 2013).

5 Results

5.1 Paragenetic sequence

Based on the mineral assemblage, texture, and the spatial distribution

relationship of ore bodies, the metallogenic process of the Huangshaping Zn-Pb polymetallic deposit can be divided into three main metallogenic periods: the skarn period, the oxide period, and the sulfide period (Fig. 4). The sulfide period is further subdivided into three metallogenic sub-stages. Twenty sphalerite samples were selected for further compositional studies. The detailed descriptive features, mineral assemblages, and ore textures are summarized in Table 1.

Skarn Period: This period develops within a range of about 150 m around the edges of granite porphyry and quartz porphyry. Skarn minerals are formed by the interaction between granite porphyry and the limestone of the Shidengzi Formation in the Lower Devonian. It can be divided into the anhydrous skarn stage and the hydrous skarn stage. The anhydrous skarn stage is dominated by garnet (Fig. 5a), while the hydrous skarn stage is characterized by a combination of diopside, amphibole, and tremolite (Fig. 5b, d).

Oxide Period: This period develops on the periphery of the hydrous skarn and is characterized by potassium feldspar, quartz, magnetite, molybdenite, and scheelite (Fig. 5e, f). Quartz is observed to be distributed in a reticular-vein form, precipitating in the skarn in the way of fissure filling and replacement, forming a massive white quartz veinlet texture. Fluorite and molybdenite can be seen in the quartz veins, which penetrate through magnetite ores and skarns, accompanied by extensive

sericitization and silicification.

Sulfide Period: It is mainly concentrated in the marbleized limestone in the contact zone of the rock mass and its adjacent crystalline limestone, within a range of about 100 to 500 m from the intrusion (Ding et al., 2016). The main mineral types include arsenopyrite, marmatite, sphalerite, pyrite galena, etc. The ore texture is mainly compact massive and shows strong lead-zinc mineralization characteristics. It can be further divided into early, middle, and late sub-stages.

Early Sulfide Stage: This sub-stage is the main lead-zinc mineralization stage, marked by euhedral-subhedral coarse-grained pyrite (Py1), euhedral acicular arsenopyrite, quartz, and fluorite, as well as sphalerite1 (Sp1; Fig. 6a).

(i) Sp1, distributed in the -176 m of the Huangshaping deposit, and a small part is distributed in the -136 m (Table 1). It shows euhedral-subhedral grains (Fig. 6a, d), with a particle size of 200–400 μm , is opaque to dark brown.

Middle Sulfide Stage: This sub-stage is dominated by subhedral coarse-grained pyrite (Py2), sphalerite2 (Sp2), fluorite, and galena, as well as a small amount of chalcopyrite (Fig. 6d). A small number of Sp2b develop the “chalcopyrite disease” texture (Fig. 6d, f). According to the mineral contact relationships and changes in mineral assemblages, two subtypes of Sp2, namely Sp2a and Sp2b, are identified.

(ii) Sp2a distributed in the 92 m of the Huangshaping deposit and a small amount is distributed in the -136 m (Table 1). It is associated with Py2, galena, and fluorite (Fig. 6d), and shows a euhedral-subhedral banded texture, with a particle size of 50–150 μm , and some develop on the periphery of Sp1, being dark brown to light brown under plane-polarized light (Fig. 6e).

(iii) Sp2b has a subhedral fine-grained texture, with a particle size of about 40–50 μm , is associated with galena, fluorite, and Py2 (Fig. 6g), and is distributed at a depth of 230 m from the surface (Table 1). It is light brown under plane-polarized light (Fig. 6g, h).

Late Sulfide Stage: This sub-stage is dominated by euhedral fine-grained pyrite³ (Py3), fluorite, calcite, euhedral-subhedral galena, bornite, and dolomite, as well as a small amount of gold-silver minerals.

5.2 Major element compositions by EPMA

A total of 32 spots of EPMA analyses were carried out on the three types of sphalerites in the Huangshaping Zn-Pb deposit, including 23 spots for Sp1, 9 spots for Sp2a, and 10 spots for Sp2b. The relevant results are shown in Appendix Table 2.

As a major element, the content of element S in the three types of sphalerites is similar, ranging from 33.25 to 33.91 wt.%. The content of Zn shows a tendency to gradually increase from the early stage, with Sp1 (mean = 49.62 wt.%), Sp2a (mean = 51.59 wt.%), and Sp2b (52.91 wt.%),

and the overall range varies from 47.79 to 53.42 wt.%. On the contrary, Fe's content tends to decrease from the early stage gradually. Sp1 has the highest content and can almost be regarded as marmatite (mean = 14.78 wt.%), followed by Sp2a (mean = 13.07 wt.%) and Sp2b (11.36 wt.%), with the overall range varying from 17.31 to 9.18 wt.%.

It is worth noting that the content of In was only detected in the Sp1 sphalerite, with an average value of 0.03 wt.%. In addition, the content of Cu fluctuates, with the highest in Sp1 (mean = 0.69 wt.%), the lowest in Sp2a (mean = 0.05 wt.%), and then an increase in Sp2b (0.21 wt.%). Mn and Sn also show similar fluctuating changes. For Sp1, Mn (mean = 0.38 wt.%) and Sn (mean = 0.11 wt.%); for Sp2a, Mn (average value = 0.03 wt.%) and Sn (below the detection limit); for Sp2b, Sn (mean = 0.29 wt.%) and Sn (mean = 0.12 wt.%).

The content of Cd is the highest in Sp2a, with an average value of 0.47 wt.%. The average value of Sp1 is 0.12 wt.%, and that of Sp2b is 0.32 wt.%. Overall, the Cd content in Sp2 sphalerite is higher than in Sp1 sphalerite.

5.3 Trace element compositions by LA-ICP-MS spot analysis

In this study, trace element tests and analyses were carried out on the three types of sphalerites in the Huangshaping lead-zinc mine by LA-ICP-MS. A total of 169 spots were analyzed. The relevant results are shown in Fig. 7 and the detailed results can be found in Appendix Table 3. Among

them, 107 spots were tested for Sp1, 42 spots for Sp2a, and 20 spots for Sp2b. In addition, to intuitively identify the testing situation of trace elements, Fig. 8 shows the corresponding representative laser output signal of the three types of sphalerites. Considering that certain elements in the same sample vary greatly, and the research pays great attention to the content of element In in sphalerite, which can vary by two to three orders of magnitude in different samples, the median and average values are used to represent the trace element compositions of the three types of sphalerite.

5.3.1 Sp1 compositions

Signals of some elements (such as In, Sn, Cu, Mn, S, Cd, and Fe) can be detected in the LA-ICP-MS output signal diagram. Their signal shows a relatively flat pattern parallel to the Zn signal (Fig. 8a). Sp1 contains some obviously enriched trace elements, such as Fe (median = 116872 ppm, mean = 75955 ppm), Mn (2428 ppm, 3723 ppm) and Cu (860 ppm, 2662 ppm). A few elements have relatively large maximum values, but the median and average values are relatively small. For example, for Ga, the maximum value is 442 ppm, the median is 7.87 ppm, and the average value is 29.46 ppm. The contents of other elements such as As, Sb, Hg, Co, Ge, and Bi are all below 30 ppm (Fig. 7). Sp1 is the richest type of sphalerite in In among the three types of sphalerite, with the highest In concentration (the maximum content is 560 ppm, median=66.31 ppm, mean=134.54 ppm) (Fig. 7; Appendix Table 3).

5.3.2 Sp2a compositions

In the LA-ICP-MS output signal diagram of Sp2a sphalerite, only signals of a few elements (such as Mn, S, Fe, Cu, In, and Sn) can be detected, and their signal also shows a relatively flat pattern parallel to the Zn signal (Fig. 8b). By comparing the contents of the three types of sphalerites, it is found that Sp2a has the least Pb, Ga, Sb, Hg, Co, Ge, and Bi. On the contrary, it has the largest Ag content, Ag (median = 69.33 ppm, mean = 108.31 ppm). In all 42 analysis spots of Sp2a, the In content exceeds the detection limit (the content range is 0.05 - 116.30 ppm, median 7.65 ppm, mean 28.64 ppm) (Fig. 7; Appendix Table 3), showing a gradually decreasing trend compared with Sp1.

5.3.3 Sp2b compositions

The LA-ICP-MS output signal diagrams of most trace elements (such as Mn, S, Cd, Fe, and In) in Sp2b show a flat trend of signal lines (Fig. 8c). Compared with Sp1 and Sp2a. Sp2b has the highest Pb content, reaching 51,158 ppm (median 3737 ppm, mean 6470 ppm), suggesting that galena may exist in Sp2b. The contents of Fe, Mn, Cd, and Cu in Sp2b are the lowest among the three types of sphalerites. Similarly, the In content in Sp2b is also very low, with a maximum value of 45.40 ppm (median 33.70 ppm, mean 34.43 ppm) (Fig. 7; Appendix Table 3).

5.4 LA-ICP-MS in-situ trace elemental mapping

This study chose representative indium-rich sphalerite samples

from the Huangshaping Zn-Pb deposit for LA-ICP-MS mapping. Two grains, Sp1 and Sp2a, were analyzed using different color and texture zoning. Sp1 has a simple opaque texture and a later Sp2b has a complex banded texture on its periphery. LA-ICP-MS mapping showed element distribution heterogeneity. In Sp1, Cd is evenly distributed (Fig. 9), Fe and Mn are in dark-brown areas, In is anomalously enriched, and Cu and Sn are in a ring-like band at the Sp1-Sp2b interface. Ga has a similar pattern but low content. In Sp2a, Fe and Cd are uniform, Mn is in dark-brown areas, In, Cu, and Sn are in dark-brown areas and absent in light ones, while Ga is in light-colored areas and scarce in dark-brown ones (Fig. 10). The different In distribution patterns in Fig. 9 and 10, as described, reflect different geological processes and physicochemical conditions during sphalerite formation. The regular, banded In distribution in Fig. 10 indicates a pulsed enrichment mechanism, which is associated with episodic influxes of hydrothermal fluid. The main controlling factors include oscillatory zoning during crystal growth, tectonically driven fluid pulses, and self-organized chemical gradients (Chouinard et al., 2005; Wu et al., 2019; Qiu et al., 2024). In contrast, the mottled texture in Fig. 9 results from post-formation processes, with the main controlling factors being partial re-equilibration, microscale heterogeneity, and deformation - induced recrystallization (Cugerone et al., 2020; Huijsmans et al., 2022; Fougereuse et al., 2023. Yang et al., 2024). The presence of both textures

within the same deposit implies multiple stages of evolution. The primary banding (Fig. 10) suggests that the main ore - forming event was dynamic and pulsed, while the mottled texture (Fig. 9) indicates that subsequent hydrothermal or metamorphic events modified the mineral. This highlights the significance of post-deposition history in determining element distribution.

6 Discussion

6.1 Substitution mechanism of trace elements in sphalerite

Sphalerite is an important host for various dispersed elements such as In, Sn, Cu, Ge, Ga, and Ag, and has been extensively studied for its occurrence states and substitution mechanisms (Cook et al., 2009; Ye et al., 2011; Belissont et al., 2014; Bonnet et al., 2016; Zheng et al., 2023; Wu et al., 2023). Three main substitution mechanisms for trace elements in sphalerite exist: (1) individual micro-mineral inclusions, (2) stoichiometric direct substitution, and (3) coupled substitution within the sphalerite lattice (Tan et al., 2024; Shu et al., 2024; Chen et al., 2024). This study investigated the correlations/substitutions among elements in sphalerite from the Huangshaping Zn-Pb polymetallic deposit. In all three types of sphalerites, Zn shows an obvious negative correlation with Fe, Mn and Cd (Fig. 11a). The LA-ICP-MS signal spikes in the laser ablation profiles are interpreted as reflecting the existence of micro or nano inclusions. In contrast, the smooth profiles indicate that elements exist in

the lattice in the form of solid solutions (Large et al., 2009). The smooth signal curves of Mn, Fe and Cd (Fig. 11) suggest that these elements directly enter the sphalerite lattice and substitute for Zn^{2+} . This is consistent with the results of previous studies (Fe^{2+} , Mn^{2+} , $\text{Cd}^{2+} \leftrightarrow \text{Zn}^{2+}$) (Pfaff et al., 2011; Torró et al., 2022; Yang et al., 2022). Nevertheless, there is no evidence indicating that Fe^{2+} , Mn^{2+} , and Cd^{2+} inhibit each other from entering sphalerite (Zhou et al., 2021). Elements with larger ionic radii and different valence states (such as Cu, Ag, In, and Sb) enter sphalerite through coupling mechanisms (Cook et al., 2009; Ye et al., 2011).

The element coupling substitution mechanisms of $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ and $(\text{Cu}, \text{Ag})^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ have been studied in detail (Cook et al., 2009; Xu et al., 2021). In the three types of sphalerites in the Huangshaping deposit, Cu-(Sn+In) shows an obvious positive correlation, with a correlation coefficient R^2 as high as 0.96 (Fig. 11b). Meanwhile, for the sphalerite Sp1 with relatively high indium content and part of Sp2a, there are also obvious positive correlations, such as Cu - In (Fig. 11c, $R^2 = 0.97$) and Cu-In (Fig. 11d, $R^2 = 0.94$). Considering the generations of sphalerite in the Huangshaping deposit, it has similar characteristics to those of the Baoshan deposit. Most of the trace elements in Sp1 are plotted above the line $[(1000 * \text{In}) / \text{Zn} = 1: 1]$ (Fig. 11e), which is consistent with Sp1 having the highest indium content, followed by Sp2a and Sp2b. In addition, an obvious “indium window” (Dill et al., 2013) also appears in Fig. 11f.

6.2 Physical and chemical conditions of ore-forming fluids

1. Sphalerite mineralization temperature

The mineralization temperatures of the three types of sphalerites in the Huangshaping deposit were also calculated using the GGIMFis thermometer method proposed by Frenzel et al. (2022). The principle is that: $T(^{\circ}\text{C}) = -(54.4 \pm 7.3) \cdot PC1^* + (208 \pm 10)$ (1) with $PC1^* = \ln\left(\frac{c_{Ga}^{0.22} \cdot c_{Ge}^{0.22}}{c_{Fe}^{0.37} \cdot c_{Mn}^{0.20} \cdot c_{In}^{0.10}}\right)$ (2), where $\ln(x)$ denotes the natural logarithm (base e), and c_i is the concentration of trace element i in sphalerite, given in units of $\mu\text{g/g}$ for Ga, Ge, Mn, and indium, and in wt.% for Fe. The calculation results are shown in the Fig. 12a. The calculated crystallization temperatures of Sp1, Sp2a, and Sp2b are 316–201 $^{\circ}\text{C}$ (median = 242 $^{\circ}\text{C}$), 299–167 $^{\circ}\text{C}$ (217 $^{\circ}\text{C}$), and 238–219 $^{\circ}\text{C}$ (234 $^{\circ}\text{C}$) respectively (Fig. 12a, Appendix Table 3). The generally decreasing temperature trend is consistent with the sequence of the Baoshan Zn-Pb polymetallic deposit (Wu et al., 2023), which also confirms that indium is mainly enriched in the early high-temperature sphalerite. This trend can also be seen in the LA-ICP-MS mapping results of Sp1, which indicated that indium is enriched in the high-temperature core and depleted in the low-temperature periphery (Fig. 9). In addition, the Huangshaping deposit generally contains relatively high contents of In and Fe, and relatively low contents of Cd and Ge. Taking Sp1a as an example, its In/Ga and Zn/Cd ratios are as high as 145.17 and 149.22,

respectively, while its Zn/Fe ratio is as low as 30.50 (calculated using the median value). Quantitatively, the calculated In/Ga, Zn/Cd and Zn/Fe ratios also indicate that the sphalerite in the Huangshaping deposit was formed under relatively high temperature conditions (mean=242 °C).

2. Sphalerite mineralization sulfur fugacity

Combined with the temperatures calculated by the GGIMFis thermometer, the sulfur fugacity during the formation of sphalerite in the Huangshaping deposit can be estimated. The calculation method is mentioned by Frenzel et al. (2022). $\log_{10}[FeS(sp)] = (0.80 \pm 0.02) \cdot$

$\log_{10}(53.1 \cdot \frac{FeS}{FeS_{max}}) - (1.38 \pm 0.03)$ (3), where FeS(sp) is the Fe-content in sphalerite (mol.% FeS), and FeS_{max} represents the temperature- and pressure-dependent maximum solubility of FeS in sphalerite when in contact with metallic Fe and troilite, described by: $FeS_{max} = 44.09 +$

$0.0125 \cdot T - \frac{3600}{T} \cdot p$ (4), where T is temperature in Kelvin, and p (0.2 ± 0.1 kbar) is pressure in kbar (Barton and Toulmin 1966; Balabin and Urusov 1995; Voudouris et al. 2008). Combined the above two equations, the

sulfur fugacity can be expressed as: $\log_{10}fS_2 = -2\log_{10}[FeS(sp)] - \frac{(1.50 \pm 0.02) \times 10^4}{T} + (14.5 \pm 0.03)$ (5). The calculation results are shown in Fig.

12b. The estimated $\log fS_2$ values gradually decrease across specimens, ranging from -13.42 to -7.11 for Sp1, -13.90 to -7.9 for Sp2a, and -12.03 to 10.90 for Sp2b. (Appendix Table 3), showing a gradual decrease in

sulfur fugacity from the early to the late stage.

3. LogfO₂-pH thermodynamic calculation

The thermodynamic analysis was carried out based on the detailed petrographic results of the Huangshaping lead-zinc deposit and the temperature calculated by the trace elements in sphalerite. The study revealed that in the early sulfide stage, closely related to the mineralization of indium-rich Sp1, the mineralization temperature was approximately 300°C. In contrast, the indium-poor Sp2a and Sp2b were formed in the middle sulfide stage, with the temperature dropping to around 200°C. The results of thermodynamic calculations are shown in Fig. 12c and d. In the Huangshaping Zn-Pb deposit, hydrogen sulfide (H₂S) is the dominant sulfur species. Zinc and iron exist in the forms of zinc ions (Zn²⁺) and ferrous ions (Fe²⁺; Qi, 2012). Under higher pH values and oxygen fugacity conditions, the indium-rich sphalerite Sp1 precipitates, while iron, arsenic and copper precipitate in the forms of magnetite or hematite, arsenopyrite and chalcopyrite respectively (Fig. 12c). On the contrary, under lower pH values and oxygen fugacity conditions, the indium-poor sphalerites Sp2a, Sp2b and pyrite precipitate (Fig. 12d). As the temperature decreased, the H₂S stability fields expand significantly, leading to the corresponding expansion of the stability fields of sphalerite, galena and pyrite. Meanwhile, the stability fields of chalcopyrite shrinks and its precipitation amount decreases accordingly. Under weakly acidic pH values and relatively high

oxygen fugacity conditions, the indium-rich sphalerite, arsenopyrite and chalcopyrite can precipitate together, and this finding is consistent with the results of petrographic studies. Moreover, calcite (CaCO_3) was found in the S-type inclusions revealed by fluid inclusion studies (Huang et al., 2013), further proving the dominant position of H_2S in the ore-forming fluid. Based on this, it can be inferred that the oxygen fugacity and pH conditions in the sulfide stage of the Huangshaping Zn-Pb deposit should be limited to the H_2S stability fields, that is, the ore-forming fluid is a reductive ore-forming fluid (Fig. 12c, d).

6.3 Metallogenic model

Indium has garnered significant attention in mineral resource research due to its unique properties and wide range of applications (Werner et al., 2017). Extensive chronological and geochemical trace element data indicate that around 175 Ma, the Izanagi plate in the southern SCB began to subduct northwestward, significantly impacting the SCB and leading to increased crustal thickness (Mao et al., 2013; Zhao et al., 2017). Between 160 and 150 Ma, changes in the internal structure of the subducting plate, such as the formation of cracks or windows, provided pathways for the ascent of shallow magmas, resulting in the formation of granitic rocks observed in the Huangshaping deposit (Fig. 13b; Wang et al., 2003). These geological changes also supplied important sources of non-ferrous metallic elements to the skarn systems in the NMB during the Yanshanian (Mao et

al., 2013; Zhao et al., 2017; Yu et al., 2022; Huang et al., 2023). The Huangshaping Zn-Pb deposit, located in the central NMB, was also affected by magma mixing with crustal materials around 160 Ma, with ascending magmatic hydrothermal fluids intruding into the marine carbonate strata of the area (Yao et al., 2006). This intrusion ultimately led to the formation of granitic rocks and skarns (Fig. 13b; Liu et al., 2022). The NNE-trending ore-controlling structures acted as “pipelines” during mineralization, controlling the migration paths and directions of ore-forming fluids (Zhai et al., 2004; Zheng, 2022). Indium is a highly volatile chalcophile element, showing moderate to high incompatibility during mantle melting (Datta et al., 2009; Li et al., 2020). During magma fractionation, most indium tends to remain in the melt (Sun, 1982), and its fractionation evolution may be similar to tin (Kovalenker et al., 1993). The Huangshaping Zn-Pb polymetallic deposit exhibits clear mineralized alteration zoning, with skarn formed in the contact zone between the surrounding rocks and the rock body. High-temperature minerals such as molybdenite and chalcopyrite develop near the rock body, while low-temperature minerals such as sphalerite and galena develop near the sedimentary rocks, typical of skarn-type Zn-Pb deposits (Einaudi & Burt, 1982; Meinert et al., 2005). As the temperature decreased and acidity increased, along with a reduction in oxygen fugacity, opaque, indium-rich Sp1 crystallized within the temperature range of 316–201°C. At 299–

167°C, relatively indium-poor Sp2a with oscillatory zoning formed. As the mineralization process progressed, the mineralization temperature further decreased, ultimately forming light brown, transparent Sp2b. Furthermore, based on the analytical results of this study, the reserve estimation method for critical metal elements mentioned by Wu et al. (2023) was utilized to estimate the reserves of the Huangshaping Zn-Pb polymetallic deposit. With the Zn reserve data referencing over 2 million tonnes (Yuan, 2015), the estimated indium resource was approximately 24 tonnes.

In this study, by employing a variety of research techniques, such as estimating the mineralization temperature of sphalerite, sulfur fugacity, and conducting thermodynamic simulations, we propose that the indium-rich ore-forming fluid is of a reductive character (Huston & Bastrakov, 2024). The indium-rich sphalerite tends to precipitate under conditions of moderate to high temperatures, elevated sulfur fugacity, high oxygen fugacity, and weak acidity. The characteristics of this ore-forming fluid are shaped by the combined influence of multiple geological factors. During the process of indium enrichment, these factors interact and mutually constrain one another. Eventually, they play a more pivotal role in determining the degree of indium enrichment within the sphalerite.

7 Conclusion

(1) Petrography has confirmed three main metallogenic stages: skarn period, oxide period, and sulfide period. During the sulfide period, three

different generations of indium-bearing sphalerite were identified: Sp1 in the early sulfide stage and Sp2a and Sp2b in the middle sulfide stage.

(2) Electron microprobe analysis, LA-ICP-MS spot analysis, and LA-ICP-MS mapping of three types of sphalerites revealed a decreasing trend in indium content from early to late stages. The earliest Sp1 has the highest indium content, with an average of 134.54 ppm and a high value of up to 560 ppm, while the indium content of the late-stage Sp2 is significantly low.

(3) The corresponding temperatures of the three types of sphalerites calculated using the GGIMFis thermometer also show a similar downward trend, from 316 °C in the early stage to 219 °C in the middle stage.

(4) Thermodynamic calculations show indium-rich sphalerite precipitates at pH of 4.5–6.0, high oxygen fugacity ($\log f_{O_2} = -32.2$ to -29.7), and high sulfur fugacity ($\log f_{S_2} = -13.43$ to -7.11). Also, $\log f_{O_2}$ and $\log f_{S_2}$ decrease from Sp1 to Sp2a and then to Sp2b.

Acknowledgments

This study is jointly funded by the National Natural Science Foundation of China (924623064, 42230807, 42022020), the National Key Research and Development Program of China (2021YFC2900300), and the Guangdong Province Introduced R&D Team (2021ZT09H399). Drs. Chengming Wang and Xiaohan Dong of the Southern Laboratory of Ocean Science and Engineering appreciated for their guidance on sample

processing. We are grateful to Editor-in-Chief Meifu Zhou and two anonymous reviewers for their insightful comments.

Journal Pre-proofs

Reference

- Belissont, R., Boiron, M.C., Luais, B., Cathelineau, M., 2014. LA-ICP-MS analyses of minor and trace elements and bulk Ge isotopes in zoned Ge-rich sphalerites from the Noailhac-Saint-Salvy deposit (France): insights into incorporation mechanisms and ore deposition processes. *Geochim. Cosmochim. Acta.* 126, 518–540.
- Berman, R.G., Brown, T.H., 1985. Heat capacity of minerals in the system $\text{Na}_2\text{O}-\text{K}_2\text{O}-\text{CaO}-\text{MgO}-\text{FeO}-\text{Fe}_2\text{O}_3-\text{Al}_2\text{O}_3-\text{SiO}_2-\text{TiO}_2-\text{H}_2\text{O}-\text{CO}_2$: representation, estimation, and high temperature extrapolation. *Contrib. Miner. Petrol.* 89, 168–183.
- Bonnet, J., Mosser-Ruck, R., Caumon, M., Rouer, O., Andre-Mater, A., Cauzid, J., Peiffert, C., 2016. Trace element distribution (Cu, Ga, Ge, Cd, and Fe) in sphalerite from the Tennessee MVT deposits, USA, by combined EMPA, LA-ICP-MS, Raman Spectroscopy, and Crystallography. *Can. Miner.* 54 (5), 1261–1284.
- Chen, X., Wang, Z.K., Yu, P.P., Wu, Y.H., Huang, Y., Tan, Z.J., Zheng, Y., 2024. Physicochemical condition of the Dafang skarn Au-Ag-Pb-Zn polymetallic deposit in the Nanling Metallogenic Belt (South China), a perspective from sphalerite mineralogy and geochemistry. *Ore Geol. Rev.* 165, 105923.
- Chu, G.B., Chen, H.Y., Zhang, S.T., Zhang, Y., Cheng, J.M., 2023. Geochemistry and geochronology of multi-generation garnet: new insights on the genesis and fluid evolution of prograde skarn formation. *Geosci. Front.* 14 (1), 101495.
- Chouinard, A., Paquette, J., Williams-Jones, A.E., 2005. Crystallographic controls on trace-element incorporation in auriferous pyrite from the Pascua epithermal high-sulfidation deposit, Chile – Argentina. *Can. Mineral.* 43(3), 951-963.

- Cook, N.J., Ciobanu, C.L., Pring, A., Skinner, W., Shimizu, M., Danyushevsky, L., Saini- Eidukat, B., Melcher, F., 2009. Trace and minor elements in sphalerite: a LA-ICPMS study. *Geochim. Cosmochim. Acta* 73, 4761–4791.
- Cugerone, A., Cenko-Tok, B., Oliot, E., Muñoz, M., Barou, F., Motto-Ros, V. Le Goff, E., 2020. Redistribution of germanium during dynamic recrystallization of sphalerite. *Geol.* 48(3), 236-241.
- Datta, A., Sinha, G., Panda, S.K., Patra, A., 2009. Growth, optical, and electrical properties of In₂S₃ zigzag nanowires. *Crystal Grow. Design.* 9(1), 427–431.
- Dick, J.M., 2019. CHNOSZ: thermodynamic calculations and diagrams for geochemistry. *Front. Earth Sci.* 7, 180.
- Dill, H.G., Garrido, M.M., Melcher, F., Gomez, M.C., Weber, B., Luna, L.I., 2013. Sulfidic and non-sulfidic indium mineralization of the epithermal Au-Cu-Zn-Pb-Ag deposit San Roque (Provincia Rio Negro, SE Argentina)-with special reference to the “indium window” in zinc sulfide. *Ore Geol. Rev.* 51, 103-128.
- Ding, T., Ma, D.S., Lu, J.J., Zhang, R.Q., Xie, Y.C., 2016. Sulfur and lead isotopic compositions of granitoids and fluid inclusions in Baoshan deposit Hunan Province. *Mineral Deposit.* 35 (4), 663–676 in Chinese with English abstract.
- Ding, T., Tan, T.T., Wang, J., Ma, D.S., Lu, J.J., Zhang, R.Q., Liang, J., Wu, B., 2022. Multiple sources for the Baoshan polymetallic Cu–Mo–Pb–Zn deposit, southern Hunan Province, China: Insights from in situ LA–MC–ICP–MS sulfur isotopic compositions. *Ore Geol. Rev.* 143, 104808.
- Einaudi, M.T., Burt, D.M., 1982. Introduction-terminology, classification, and composition of skarn deposits. *Econ. Geol.* 77, 745–754.

- Frenzel, M., Hirsch, T., Gutzmer, J., 2016. Gallium, germanium, indium, and other trace and minor elements in sphalerite as a function of deposit type—a meta-analysis. *Ore Geol. Rev.* 76, 52–78.
- Frenzel, M., Mikolajczak, C., Reuter, M.A., Gutzmer, J., 2017. Quantifying the relative availability of high-tech by-product metals—the cases of gallium, germanium and indium. *Res. Policy.* 52, 327–335.
- Frenzel, M., Voudouris, P., Cook, N.J., Ciobanu, C.L., Gilbert, S., Wade, B.P., 2022. Evolution of a hydrothermal ore-forming system recorded by sulfide mineral chemistry: a case study from the Plaka Pb–Zn–Ag Deposit, Lavrion Greece. *Miner. Deposita.* 57 (3), 417–438.
- Fougereuse, D., Cugerone, A., Reddy, S.M., Luo, K. Motto-Ros, V., 2023. Nanoscale distribution of Ge in Cu-rich sphalerite. *Geochim. Cosmochim. Acta*, 346, 223–230.
- Gagnevin, D., Menuge, J.F., Kronz, A., Barrie, C., Boyce, A.J., 2014. Minor elements in layered sphalerite as a record of fluid origin, mixing, and crystallization in the Navan Pb–Zn ore deposit Ireland. *Econ. Geol.* 109, 1513–1528.
- Greaney, A.T., Rudnick, R.L., Helz, R.T., Gaschnig, R.M., Piccoli, P.M., Ash, R.D., 2017. The behavior of chalcophile elements during magmatic differentiation as observed in Kilauea Iki lava lake, Hawaii. *Geochim. Cosmochim. Acta* 210, 71–96.
- Helgeson, H.C., Kirkham, D.H., Flowers, G.C., 1981. Theoretical prediction of the thermodynamic behavior of aqueous electrolytes by high pressures and temperatures: IV. Calculation of activity coefficients, osmotic coefficients, and apparent molal and standard and relative partial molal properties to 600°C and 5kb. *Am. J. Sci.* 281, 1249–1516.

- Hong, J., Zhang, H., Li, D., Ouyang, Y., Zhai, D., Liu, F., Liu, J., 2022. In situ LA-ICP-MS U-Pb geochronology and geochemical characteristics of garnet from the Zhuxi skarn W-Cu deposit, South China. *Ore Geol. Rev.* 140, 104577.
- Huang, C., Li, X.F., Wang, L.F., Liu, F.P., 2013. Fluid inclusion study of the Huangshaping polymetallic deposit, Hunan Province, South China. *Acta Petrol. Sin.* 29(12), 4232–4244 in Chinese with English abstract.
- Huang, Y., Yu, P.P., Chen, X., Wu, Y.H., Wang, Z.K., Ding, W., Zheng, Y., 2023. Magmatic controls on Pb-Zn and W-Sn metallogenic variations in the Nanling Metallogenic Belt, South China: Difference in sources, oxygen fugacity and differentiation degree. *Ore Geol. Rev.* 162, 105701.
- Huijsmans, J.R., Hamers, M., Drury, M.R. Lee, J.K., 2022. Recrystallisation and trace-element mobility in zircons: Implications for U-Pb dating. *Minerals.* 12(12), 1489.
- Huston, D.L., Bastrakov, E., 2024. Germanium, indium, gallium and cadmium in zinc ores: a mineral system approach. *Aust. J. Earth Sci.*
- Ishihara, S., Murakami, H., 2008. Geologic background of In-resources in the southern China. *Shigen-Chishitsu.* 58, 139–150 in Japanese with English abstract.
- Ishihara, S., Murakami, H., Li, X., 2011. Indium concentration in zinc ores in plutonic and volcanic environments: Examples at the Dulong and Dachang mines. South China. *Bull. Geol. Surv. Jpn.* 62, 259–272.
- Kovalenker, V.A., Jeleň, S., Sandomirskaya, S., 1993. Minerals of the system Ag-Cu-Pb-Bi-S from the polymetallic veins of the Stiavnica-Hodrusa ore field (Slovakia). *Geologica. Carpathica.* 44(6), 409–419.

- Kumar, A.A., Sanislav, I., Cathey, H., Dirks, P., 2023. Geochemistry of indium in magmatic-hydrothermal tin and sulfide deposits of the Herberton Mineral Field, Australia. *Mineral. Depos.* 58(7), 1297–1316.
- Large, R.R., Danyushevsky, L., Hollit, C., Maslennikov, V., Meffre, S., Gilbert, S., Bull, S., Scott, R., Emsbo, P., Thomas, H., Singh, B., Foster, J., 2009. Gold and trace element zonation in pyrite using a laser imaging technique: implications for the timing of gold in orogenic and carlin-style sediment-hosted deposits. *Econ. Geol.* 104 (5), 635–668.
- Li, X.F., Zhu, Y.T., Xu, J., 2020. Indium as a critical mineral: A research progress report. *Chin Sci. Bull.* 65, 3678–3687 in Chinese with English Abstract.
- Li, X.H., Li, W.X., Li, Z.X., Lo, C.H., Wang, J., Ye, M.F., Yang, Y.H., 2009. Amalgamation between the Yangtze and Cathaysia Blocks in South China: Constraints from SHRIMP U-Pb zircon ages, geochemistry and Nd–Hf isotopes of the Shuangxiwu volcanic rocks. *Precambrian Res.* 174 (1–2), 117–128.
- Li, X.M., Zhang, Y.X., Li, Z.K., Zhao, X.F., Zuo, R.G., Xiao, F., Zheng, Y., 2023. Discrimination of Pb-Zn deposit types using sphalerite geochemistry: New insights from machine learning algorithm. *Geosci. Front.* 14 (4), 101580.
- Liu, J.P., Rong, Y.N., Zhang, S.G., Liu, Z.F., Chen, W.K., 2017. Indium Mineralization in the Xianghualing Sn-Polymetallic Orefield in Southern Hunan, Southern China. *Minerals.* 7(9), 173.
- Liu, W., Mei, Y., Etschmann, B., Glenn, M., MacRae, C., Spinks, S.C., Ryan, C.G., Brugger, J., Paterson, D.J., 2023. Germanium speciation in experimental and natural sphalerite: implications for critical metal enrichment in hydrothermal Pb-Zn ores. *Geochim. Cosmochim. Acta* 342, 198–214.

- Liu, Y.S., Hu, Z.C., Gao, S., Günther, D., Xu, J., Gao, C.G., Chen, H.H., 2008. In situ analysis of major and trace elements of anhydrous minerals by LA-ICP-MS without applying an internal standard. *Chem. Geol.* 257 (1–2), 34–43.
- Liu, Z., Qian, X., Gan, C.S., Cui, X., Zhang, Y.Z., Wang, Y., Wang, Y.J., 2022. Migration and tectonic implications of Late Jurassic mafic magmatism in South China. *J. Geol. Soc.* 179(6), 1–16.
- Lockington, J.A., Cook, N.J., Ciobanu, C.L., 2014. Trace and minor elements in sphalerite from metamorphosed sulphide deposits. *Miner. Petrol.* 108, 873–890.
- Lu, Y.Y., Fu, J.M., Tan, S.M., Cheng, S.B., Guo, Z.Y., Qin, C.W., Liu, B.D., Ma, L.Y., 2017. Zircon U-Pb dating, Hf isotopic compositions and their prospecting signification of the granodiorite porphyry from Dafang Au-Ag-Pb-Zn deposit, Southern Hunan province. *Geol. Miner. Resour. South China* 33 (3), 262–274 in Chinese with English abstract.
- Mao, J.W., Cheng, Y.B., Chen, M.H., Pirajno, F., 2013. Major types and time-space distribution of Mesozoic ore deposits in South China and their geodynamic settings. *Mineral. Depos.* 48, 267–294.
- Meinert, L.D., Dipple, G.M., Nicolescu, S., 2005. World Skarn Deposits. *Econ. Geol. Geology 100th Anniversary Volume.* 299–336.
- Paton, C., Hellstrom, J., Paul, B., Woodhead, J., Hergt, J., 2011. Iolite: Freeware for the visualisation and processing of mass spectrometric data. *J. Anal. At. Spectrom.* 26 (12), 2508–2518.
- Pearson, R.G., 1963. Hard and soft acids and bases. *J. Am. Chem. Soc.* 85(22), 3533–3539.

- Pfaff, K., Koenig, A., Wenzel, T., Ridley, I., Hildebrandt, L.H., Leach, D.L., Markl, G., 2011. Trace and minor element variations and sulfur isotopes in crystalline and colloform ZnS: Incorporation mechanisms and implications for their genesis. *Chem. Geol.* 286, 118–134.
- Qi, F.Y., 2012. Magmatic-fluvial Metallogensis and Its Evolution in Baoshan Cu-Pb-Zn Polymetallic Deposit, Hunan Province. China University of Geosciences, pp. 1 – 100. MSc dissertation in Chinese.
- Rudnick, R.L., Gao, S., 2003. Composition of the continental crust. *Treatise on Geochemistry*. 3, 1–64.
- Scott, S.D., Barnes, H.L., 1971. Sphalerite geothermometry and geobarometry. *Econ. Geol.* 66, 653–669.
- Seward, T.M., Henderson, C.M.B., Charnock, J.M., 2000. Indium (III) chloride complexing and solvation in hydrothermal solutions to 350 °C: an EXAFS study. *Chem. Geol.* 167(1-2), 117–127.
- Shu, L., Charvet, J., 1996. Kinematics and geochronology of the proterozoic DongxiangShexian ductile shear zone: With HP metamorphism and ophiolitic melange (Jiangnan region, South China). *Tectonophys.* 267 (1–4), 291–302.
- Shu, L.H., Chen, X., Yu, P.P., Huang, Y., Wang, X.H., Li, R.J., Tan, Z.J., Chen, M.X., Zheng, Y., 2024. Oxidative fluids facilitate Cd enrichment in a magmatic-hydrothermal system: An example from the Kangjiawan Zn-Pb polymetallic deposit, South China. *Ore Geol. Rev.* 165, 105926.
- Shu, X.J., Wang, X.L., Sun, T., Xu, X.S., Dai, M.N., 2011. Trace elements, U-Pb ages and Hf isotopes of zircons from Mesozoic granites in the western Nanling Range, South China: implications for petrogenesis and W-Sn mineralization. *Lithos.* 127, 468–482.

- Tan, Z.J, Zheng, Y., Yu, P.P., Li, R.J, Huang, Y., Ding, W., Wu, Y.H, Chen, M.X., Sun, L.H, Wang, Z.K., Chen, X., 2024. Indium concentrations as a potential indicator of orebody-intrusion distance in a skarn Zn-Pb system, elucidated by the Fozichong Orefield (South China). *Ore Geol. Rev.* 165, 105912.
- Torró, L., Benites, D., Vallance, J.T., Laurent, O., Ortiz-Benavente, B.A., Chelle-Michou, C., Proenza, J.A., Fontboté, L., 2022. Trace element geochemistry of sphalerite and chalcopyrite in arc-hosted VMS deposits. *J. Geochem. Explor.* 232, 106882.
- USGS (United States Geological Survey), 2021. Mineral Commodity Summaries 2011-2021. <https://www.usgs.gov/centers/nmic/indium-statistics-and-information>.
- Wang, D., Chen, Y., Chen, Z., Liu, S., Xu, J., Zhang, J., Zeng, Z., Chen, F., Li, H., Guo, C., 2007. Assessment on mineral resource in Nanling Region and suggestion for further prospecting. *Acta Geol. Sin.* 81, 882–890.
- Wang, K., Zhai, D., Williams-Jones, A.E., Li, D., Liu, J., 2023. Discrete late Jurassic Sn mineralizing events in the Xianghualing Ore District, South China: Constraints from cassiterite and garnet U-Pb geochronology. *Am. Mineral* 108 (7), 1384–1398.
- Wang, X.L., Zhou, J.C., Griffin, W.L., Zhao, G., Yu, J.H., Qiu, J.S., Zhang, Y.J., Xing, G.F., 2014. Geochemical zonation across a Neoproterozoic orogenic belt: Isotopic evidence from granitoids and metasedimentary rocks of the Jiangnan orogen, China. *Precambrian Res.* 242, 154–171.
- Wang, Y.J., Fan, W.M., Guo, F., 2003. Geochemistry of early Mesozoic potassium-rich diorites-granodiorites in southeastern Hunan Province, South China: Petrogenesis and tectonic implications. *Geochem. J.* 37, 427–448.

- Werner, T.T., Mudd, G.M., Jowitt, S.M., 2017. The world's by-product and critical metal resources part III: a global assessment of indium. *Ore Geol. Rev.* 86, 939–956.
- Wilson, S.A., Koenig, A.E., Ridley, W.I., 2002. Development of sulfide calibration standards for the laser ablation inductively-coupled plasma mass spectrometry. *J. Anal. At. Spectrom.* 17, 406–409.
- Wood, S.A., Samson, I.M., 2006. The aqueous geochemistry of gallium, germanium, indium and scandium. *Ore Geol. Rev.* 28(1), 57–102.
- Wu, Y.F., Fougereuse, D., Evans, K., Reddy, S.M., Saxey, D.W., Guagliardo, P. Li, J.W., 2019. Gold, arsenic, and copper zoning in pyrite: A record of fluid chemistry and growth kinetics. *Geol.* 47(7), 641–644.
- Wu, Y.H., Yu, P.P., Chen, X., Hu, Z.B., Long, L.J., He, B., Zheng, Y., 2023. Earlier stage, higher temperature, and deeper space facilitate indium precipitation in a skarn system, as exemplified by the Baoshan Zn-Pb polymetallic deposit, South China. *Ore Geol. Rev.* 163, 105745.
- Xu, J., Ciobanu, C.L., Cook, N.J., Slattery, A.D., Kontonikas-Charos, A., 2021. Phase relationships in the system ZnS-CuInS_2 : insights from a nanoscale study of indiumbearing sphalerite. *Am. Mineral.* 106 (2), 192–205.
- Yang, L., Wang, Q.F., Fougereuse, D., Xian, H.Y., Li, H.J., Wang, R.X., Rickard, W.D., Groves, D.I. Deng, J., 2024. Release and re-enrichment of invisible gold in arsenian pyrite promoted by coupled dissolution-reprecipitation reactions. *Commun. Earth Environ.* 5(1), 241.

- Yang, Q., Zhang, X.J., Ulrich, T., Zhang, J., Wang, J., 2022. Trace element compositions of sulfides from Pb-Zn deposits in the Northeast Yunnan and northwest Guizhou Provinces, SW China: insights from LA-ICP-MS analyses of sphalerite and pyrite. *Ore Geol. Rev.* 141, 104639.
- Yao, J.M., Hua, R.M., Lin, J.F., 2006. REE, Pb-S isotope geochemistry and Rb-Sr isochron age of pyrites in the Baoshan deposit, South Hunan Province China. *Acta Geol. Sin.* 80(7), 1045–1054 in Chinese with English abstract.
- Ye, L., Cook, N.J., Ciobanu, C.L., Liu, Y.P., Zhang, Q., Liu, T.G., Gao, W., Yang, Y.L., Danyushevskiy, L., 2011. Trace and minor elements in sphalerite from base metal deposits in South China: a LA-ICPMS study. *Ore Geol. Rev.* 39, 188–217.
- Yi, W., Halliday, A.N., Alt, J.C., Lee, D.C., Rehkämper, M., Garcia, M.O., Su, Y., 2000. Cadmium, indium, tin, tellurium, and sulfur in oceanic basalts: implications for chalcophile element fractionation in the Earth. *J. Geophys. Res. Solid Earth* 105(B8), 18927–18948.
- Yu, P.P., Zheng, Y., Cawood, P.A., Wu, Y.H., Hu, Z.B., Zhang, Y.Z., 2022. Setting and formation of the earliest Neoproterozoic rifted arc Pingshui VMS deposit. South China. *Precambrian Res.* 369, 106548.
- Yu, P.P., Ding, W., Zeng, C.Y., Liu, Y., Zhou, Y.Z., Zheng, Y., 2023. Episodic Magmatism and Continental Reworking in the Yunkai Domain, South China. *Earth Sci.* 48(9), 3205–3220 in Chinese with English abstract.
- Yuan, S.D., Peng, J.T., Hao, S., Li, H.M., Geng, J.Z., Zhang, D.L., 2011. In situ LA-ICP-MS and ID-TIMS U-Pb geochronology of cassiterite in the giant Furong tin deposit, Hunan Province, South China: new constraints on the timing of tin-polymetallic mineralization. *Ore Geol. Rev.* 43, 235–242.

- Yuan, Y.B., 2015. The Genetic Difference Between Two Metallogenic Granites and the Ore-forming Material Source of the Huangshaping Deposit in Southern Hunan. China University of Geosciences (Beijing), Beijing, China in Chinese.
- Zhai, Y.S., Peng, R.M., Xiang, Y.C., Wang, J.P., Deng, J., You, Z.D., Wang, Y.H., Qi, K.J., 2004. Methodology of Regional Metallogenic Research. China Land Press, Beijing in Chinese.
- Zhang, Q., 1987. Trace elements in galena and sphalerite and their geochemical significance in distinguishing the genetic types of Pb-Zn ore deposits. *Chin. J. Geochem.* 6, 177–190.
- Zhang, Q., Zhan, X.Z., Pan, J.Y., Shao, S.X., 1998. Geochemical enrichment and mineralization of indium. *Chin. J. Geochem.* 17, 221–225.
- Zhang, Z., 2015. Research on metallization of the Huangshaping tungsten polymetallic deposit, Hunan province, China. China University of Geosciences (Beijing), Beijing, China in Chinese.
- Zhao, F., 2017. Geological characteristics and the mineralization process from Huangshaping W-Mo polymetallic deposit in Hunan province. China University of Geosciences (Beijing), Beijing, China in Chinese.
- Zhao, G., Cawood, P.A., 1999. Tectonothermal evolution of the Mayuan Assemblage in the Cathaysia Block; implications for Neoproterozoic collision-related assembly of the South China Craton. *Am. J. Sci.* 299 (4), 309–339.
- Zhao, P.L., Yuan, S.D., Mao, J.W., Santosh, M., Zhang, D.L., 2017. Zircon U-Pb and Hf-O isotopes trace the architecture of polymetallic deposits: a case study of the Jurassic ore-forming porphyries in the Qin-Hang metallogenic belt, China. *Lithos.* 292, 132–145.

- Zhao, T.P., Chen, C., He, X.H., Meng, L., Xu, J., Liu, W.Y., 2022. A synthesis of the geology, spatial-temporal distribution and enrichment mechanism of granite-related indium deposits in China. *Ore Geol. Rev.* 146, 104932.
- Zheng, Y., 2022. Large-Scaled Structure-Alteration-Mineralization Mapping of the Hydrothermal Deposits: Basic Principle and Precautions. *Earth Sci.* 47(10), 3603–3615.
- Zheng, Y., Yu, P.P., Li, Z.K., Xiong, S.F., Zhou, L.L., Zhou, J.X., Wang, C.M., Meng, Y.M., Zhang, Y., Wang, Y.J., Xu, J., Wu, Y., Guo, L.X., Zhao, T.P., 2023. Critical Metals Ga, Ge and In in the Global Zn-Pb Deposits: Current Understanding, Challenges and Perspectives. *J. Earth Sci.* 34, 1308–1311.
- Zhou, L.L., Fan, H., Ulrich, T., 2021. Editorial for Special Issue “Critical Metals in Hydrothermal Ores: Resources, Recovery, and Challenges”. *Minerals*. 11(3), 299.
- Zhou, M., Gao, J., Zhao, Z., Zhao, W.W., 2018. Introduction to the special issue of Mesozoic W-Sn deposits in South China. *Ore Geol. Rev.* 101, 432–436.
- Zhou, X.M., Sun, T., Shen, W.Z., Shu, L.S., Niu, Y.L., 2006. Petrogenesis of Mesozoic granitoids and volcanic rocks in South China: A response to tectonic evolution. *Episodes*. 29 (1), 26–33.
- Zhu, D.P., Li, H., Algeo, T.J., Jiang, W.C., Wang, C., 2021. The prograde-to-retrograde evolution of the Huangshaping skarn deposit (Nanling Range, South China). *Miner. Deposita*. 56, 1087–1110.
- Zhu, D.P., Li, H., Jiang, W.C., Wang, C., Hu, X.J., Kong, H., 2022. Ore-forming environment of Pb–Zn mineralization related to granite porphyry at Huangshaping skarn deposit, Nanling Range, South China. *Trans. Nonferrous Met. Soc. China* 32, 3015–3035.

Table Captions

Table 1. Sphalerite Sampling location at Huangshaping Zn-Pb deposit.

Figure Captions

Fig. 1. (a) Simplified tectonic map of China showing major cratonic blocks and orogenic belts (modified from Zhao et al., 2001). (b) Map showing the distribution of the skarn-type Zn-Pb deposits in South China (modified from Su et al., 2017)

Fig. 2. Simplified geologic map of the Huangshaping deposit (modified from Ding et al., 2016).

Fig. 3. (a) The cross-section of A-A' through the Huangshaping deposit. (b) The cross-section of B-B' through the Huangshaping deposit (modified from Ding et al., 2016).

Fig. 4. Alteration and mineralization paragenetic sequence at the Huangshaping deposit.

Fig. 5. Photos of hand specimens and thin sections of samples from the Huangshaping deposit. (a) Massive garnet-bearing skarn in light yellow; massive sulfide ore containing molybdenite; (b) euhedral garnet thin section (left) and image under crossed polarized light (right) with epidote; (c) diopside under single plane-polarized light (left) and crossed polarized light (right); molybdenite, garnet and diopside coexisting under crossed polarized light (d) and reflected light (e); f, magnetite and quartz with early sulfide stage chalcopyrite under reflected light. Abbreviations: Grt = garnet; Di = diopside; Qz = quartz; Ep = epidote; Fl = fluorite; Ccp = chalcopyrite; Mag = magnetite; Mo = molybdenite.

Fig. 6. Petrographic images of different types of sphalerites in the Huangshaping. (a) Sphalerite1 intergrown with pyrite1 and late stage pyrite2 under reflected light; (b) Sphalerite1 have opaque property under plane-polarized light (b); (c) Schematic diagram of sphalerite1 characteristics and mineral symbiosis; Sphalerite1, sphalerite2a with euhedral-subhedral banded texture, pyrite2, galena, and fluorite under reflected light (d) and under plane-polarized light (e); (f) Schematic diagram of sphalerite2a characteristics and mineral symbiosis; (g) Sphalerite2b have transparent property under plane-polarized light; (h) Sphalerite Sp2b with pyrite1 and chalcopyrite under plane-polarized light (left) and reflected light (right). (i) Schematic diagram of sphalerite2b characteristics and mineral symbiosis; Abbreviations: Ccp = chalcopyrite; Py = pyrite; Gn = Galena; Fl = fluorite; Qz = quartz; Sp = sphalerite; Py = pyrite; Apy = arsenopyrite; Fl = fluorite.

Fig. 7. Box and whisker plot for trace element contents of the three sphalerite types from the Huangshaping deposit.

Fig. 8. Representative LA-ICP-MS output signal for the Huangshaping sphalerite.

Fig. 9. LA-ICP-MS maps of trace elements in the sphalerite1

Fig. 10. LA-ICP-MS maps of trace elements in the sphalerite2a.

Fig. 11. Binary plots of (a) Zn vs. (Fe + Cd + Mn); (b) (Ga + In + Sn + Sb) vs. Cu; (c) In vs. Cu; (d) In vs. Sn; (e) Zn vs. (1000*In); (f) Cd vs. In for the Huangshaping deposit.

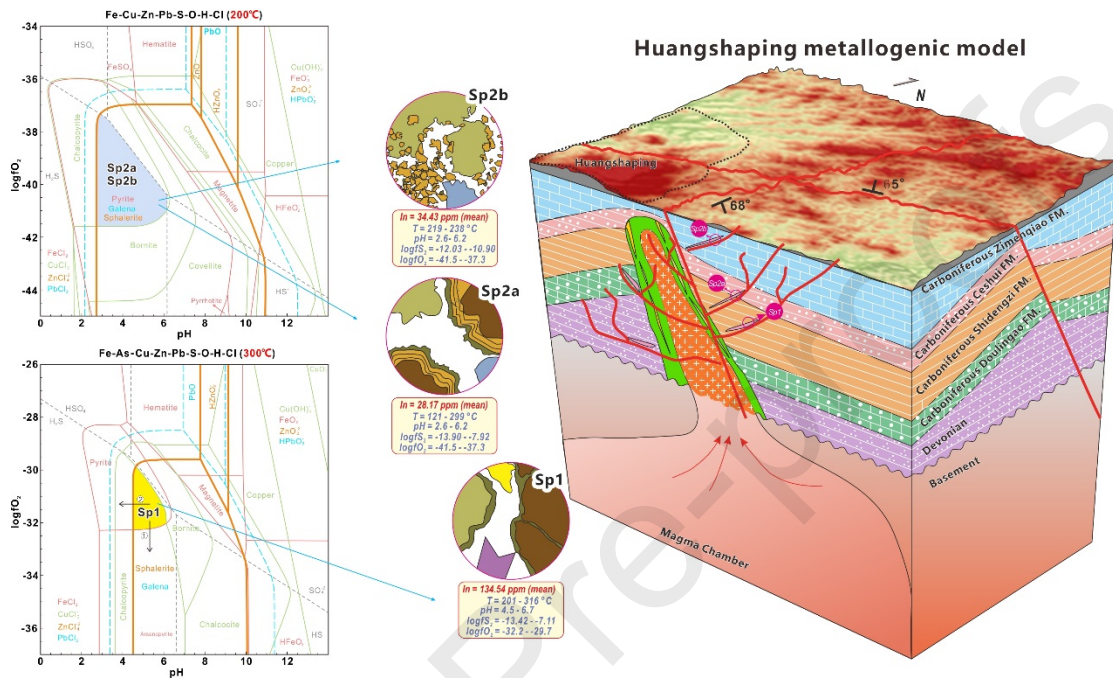
Fig. 12. (a) Box and whisker plot for indium contents and temperature range of the three types of sphalerites from the Huangshaping deposit. (b) Temperature vs. $\log fS_2$ diagram adapted from Einaudi et al. (2003). The different shadow field represents the range of the three types of sphalerite sulfur fugacity. (c) $\log fO_2$ versus pH diagrams for Fe-As-S-O-H-Cl system (red line) together with Fe-Cu-S-O-H-Cl system (green line), Pb-S-O-H-Cl system (blue dash line), Zn-S-O-H-Cl system (yellow line) and aqueous sulfur species (grey dash line) showing stability fields for Sphalerite (ZnS), Pyrite (FeS), and Arsenopyrite ($FeAsS$) calculated for 300°C (c) and 200°C. (d) corresponding to the early sulfide stage and middle sulfide stage, respectively, at the Huangshaping deposit. The yellow shade in Fig. 12c represents the range of fluid conditions during the early sulfide stage, while the blue shade in Fig. 12d shows the range of fluid conditions during the middle sulfide stage. The black arrows indicate two potential pathways for fluids, involving the decrease in oxygen fugacity (1) and the decrease in pH (2). Both diagrams calculated for $\Sigma S = 0.1$ m, $\Sigma Fe = 0.0001$ m, $\Sigma As = 0.0001$ m, $\Sigma Cu = 0.0001$ m, $\Sigma Zn = 0.0001$ m, $\Sigma Pb = 0.0001$ m, $\Sigma NaCl = 1$ m, and $P = 1000$ bar. Thermodynamic data of other species are cited from the OBIGT thermodynamic database of CHNOSZ.

Fig. 13. Schematic diagram illustrating the metallogenic model and ore-forming processes of the Huangshaping deposit. The magma from the basement intrudes into the carbonate strata of the caprock, leading to skarnization. Indium content is the mean value.

Appendix Table 2 & Table 3. Supplementary data

Appendix Table 2. Summary statistics of EPMA data for sphalerite (wt. %).

Appendix Table 3. Summary statistics of LA-ICP-MS data and temperature estimated results for sphalerite (ppm).



Highlights

- Three sphalerite generations with decreasing In concentrations were identified at the Huangshaping skarn-type Zn-Pb deposit.
- Early-stage sphalerite in the Huangshaping deposit has high indium concentrations.
- Thermodynamic calculations indicate Indium-rich sphalerite precipitates under weakly acidic pH, high oxygen fugacity, high temperature, and high sulfur fugacity in a reduced, H₂S-rich environment.

CRedit authorship contribution statement

Yihan Wu: Data Curation, Formal analysis, Writing-Original Draft. **Yi**

Zheng: Formal analysis, Funding acquisition. **Pengpeng Yu:** Conceptualization, Methodology, Writing-Review & Editing. **Yin Huang:** Investigation, Formal analysis. **Jie Yu:** Conceptualization, Methodology, Visualization. **Taiping Zhao:** Conceptualization, Visualization. **Qiuming Cheng:** Conceptualization, Validation, Formal analysis. **Xi Chen:** Supervision, Investigation, Writing-Review & Editing, Formal analysis.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Table 1. Sphalerite sampling location at Huangshaping Zn-Pb deposit

Types	Number	Sampling location	Mineral assemblages	Texture	Analytical methodology
Sp1	HSP-6	-136 m	Qz-Py-Ccp-Sp-Fl	euhedral-subhedral	LA-ICPMS
Sp1	HSP-22	-136 m	Qz-Py-Ccp-Sp-Fl	euhedral-subhedral	EPMA & LA-ICPMS
Sp1	HSP-22B g	-136 m	Qz-Py-Ccp-Sp-Fl	euhedral-subhedral	EPMA & LA-ICPMS
Sp1	HSP-30A g	-136 m	Qz-Py-Ccp-Sp-Fl	euhedral-subhedral	EPMA & LA-ICPMS
Sp1	HSP-40	-176 m	Qz-Py-Ccp-Sp-Fl	euhedral-subhedral	EPMA & LA-ICPMS
Sp1	HSP-51	-176 m	Qz-Py-Ccp-Sp-Fl	euhedral-subhedral	EPMA & LA-ICPMS

Sp1	HSP-51 AB	-176 m	Qz-Py- Ccp-Sp-Fl	euhedral- subhedral	EPMA & LA- ICPMS
Sp1	HSP-51 BB	-176 m	Qz-Py- Ccp-Sp-Fl	euhedral- subhedral	EPMA & LA- ICPMS
Sp1	HSP-51 CB	-176 m	Qz-Py- Ccp-Sp-Fl	euhedral- subhedral	EPMA & LA- ICPMS
Sp2a	HSP-26	92 m	Fl-Py- Sp-Gn	banded	EPMA & LA- ICPMS
Sp2a	HSP-30B	92 m	Fl-Py- Sp-Gn	banded	EPMA & LA- ICPMS
Sp2a	HSP-30B-1 g	92 m	Fl-Py- Sp-Gn	banded	EPMA & LA- ICPMS
Sp2a	HSP-30B B	92 m	Fl-Py- Sp-Gn	banded	EPMA & LA- ICPMS
Sp2a	HSP-31-2	92 m	Fl-Py- Sp-Gn	banded	EPMA & LA- ICPMS
Sp2b	HZJ-19	120 m	Fl-Py- Sp-Gn	subhedral fine	EPMA & LA- ICPMS

Abbreviations: Ccp = chalcopyrite; Py = pyrite; Gn = Galena; Cal = Calcite; Sp = Sphalerite;

Figure 1

[Click here to access/download;Figure;Fig.1 HSP Tectonic map2.jpg](#)

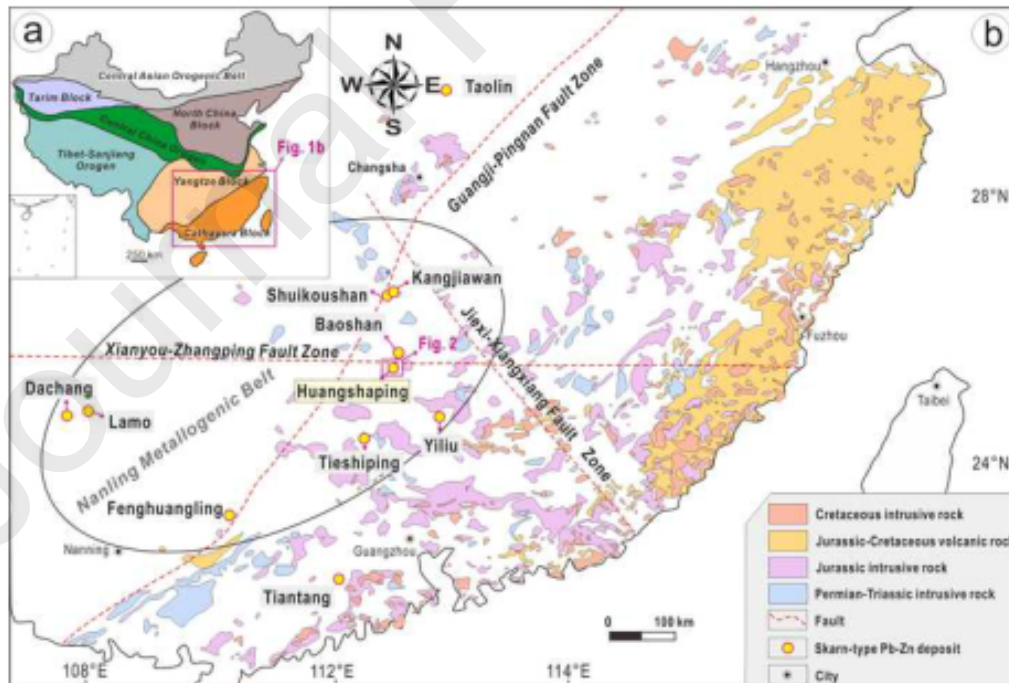


Figure2

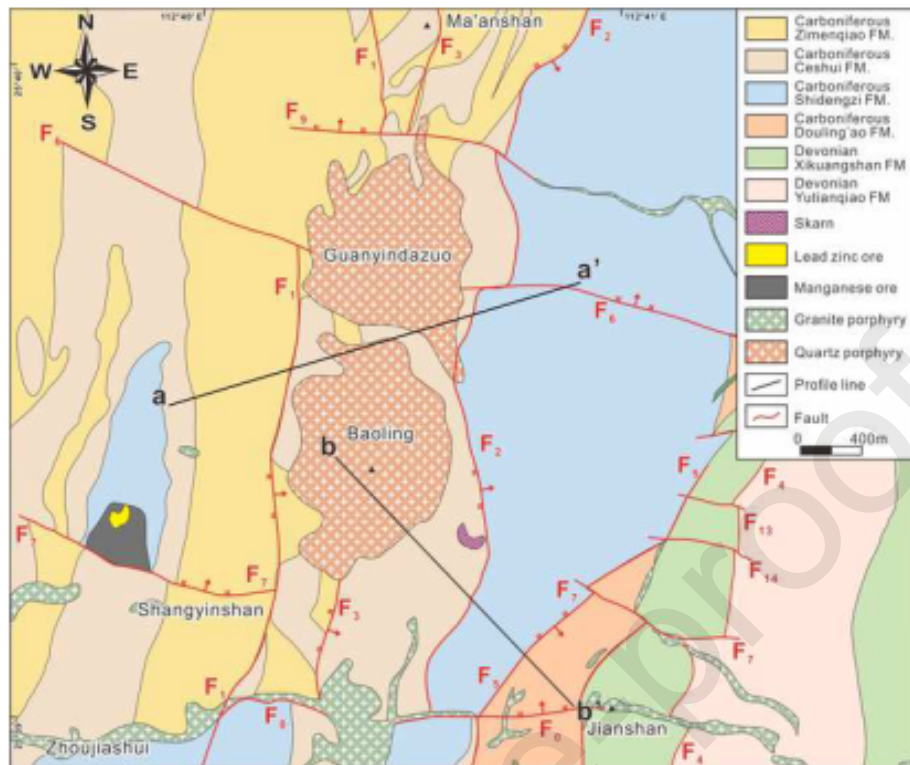
[Click here to access/download;Figure;Fig.2 HSP ore deposit geology map2.jpg](#)

Figure3

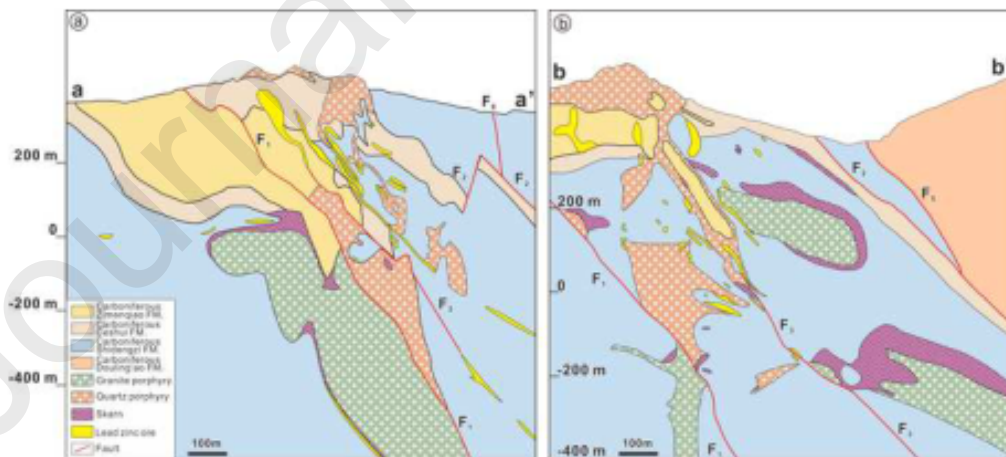
[Click here to access/download;Figure;Fig.3 HSP cross section2.jpg](#)

Figure4

[Click here to access/download;Figure;Fig.4 HSP generation sequence2.jpg](#)

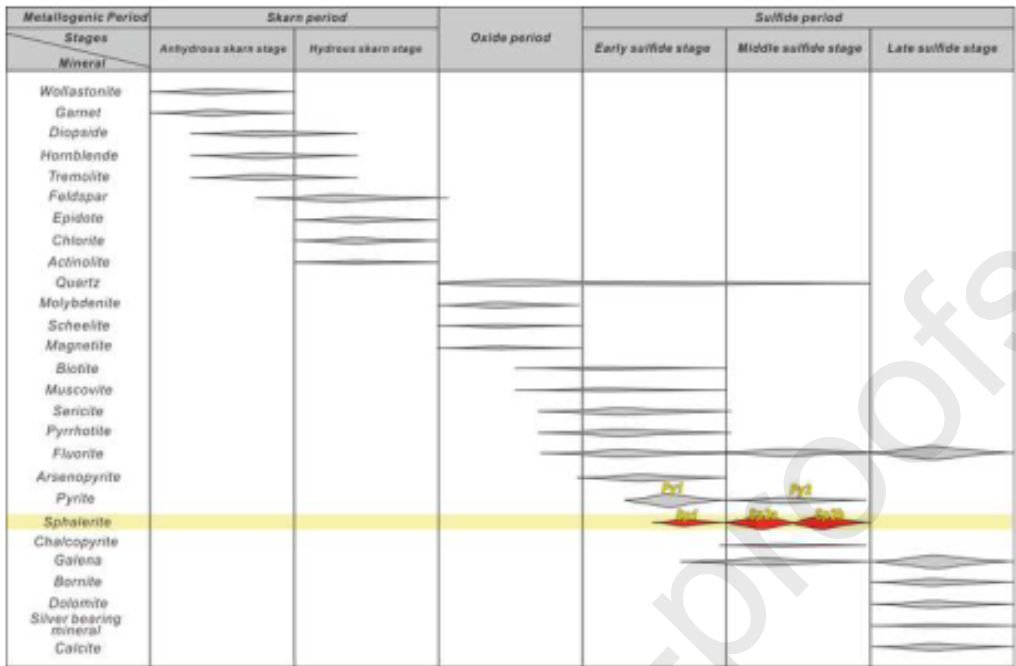


Figure5

[Click here to access/download;Figure;Fig.5 HSP skarn minerals.jpg](#)

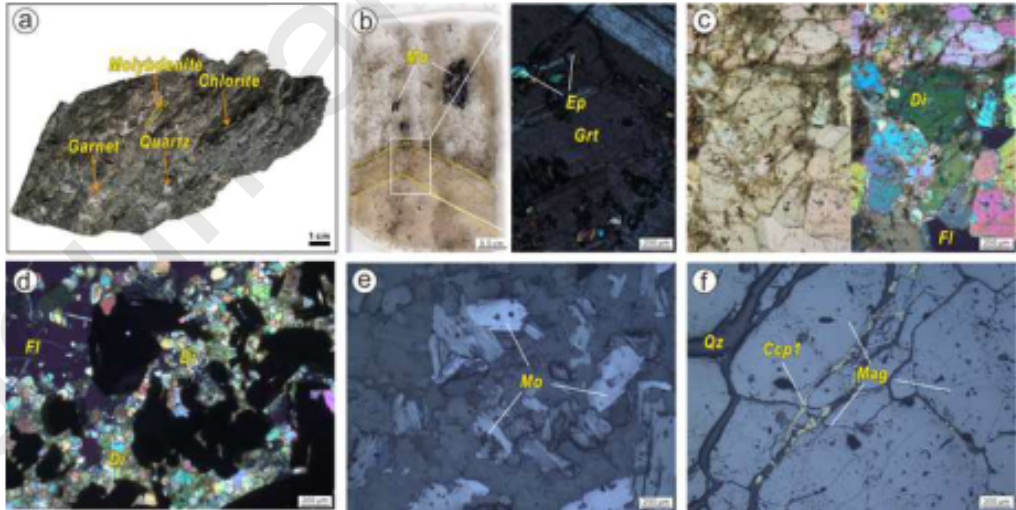


Figure6

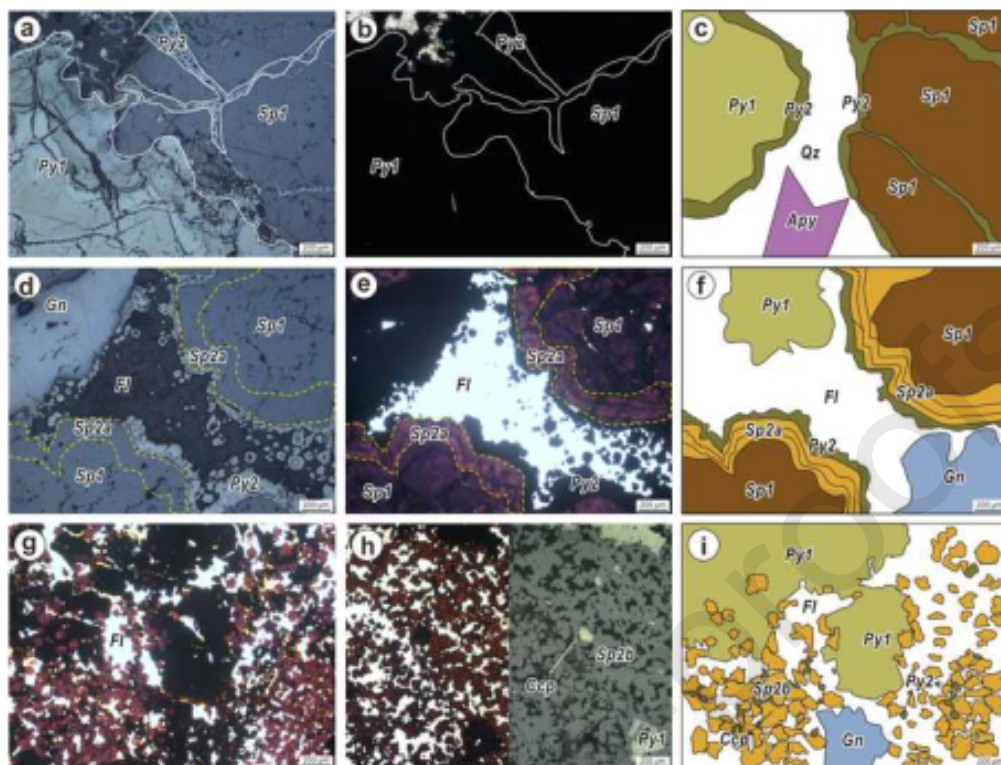
[Click here to access/download;Figure;Fig.6 HSP aphaerites.jpg](#)

Figure7

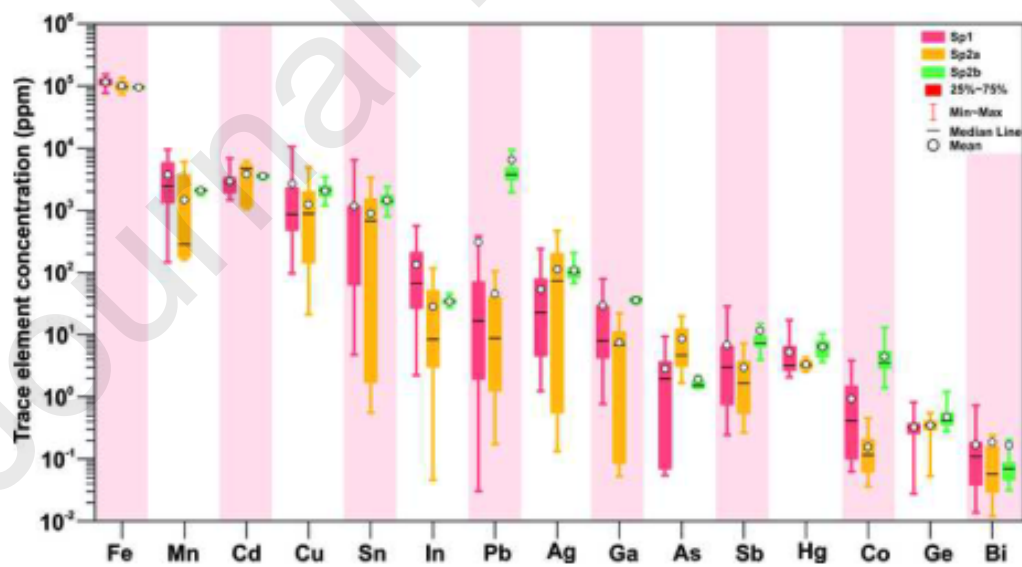
[Click here to access/download;Figure;Fig.7 HSP box and whisker plot.jpg](#)

Figure8

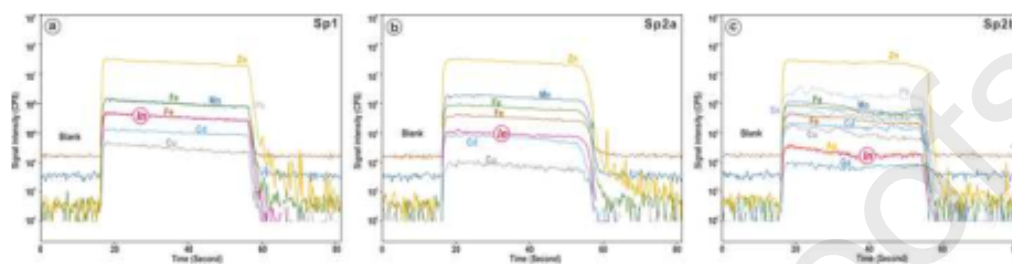
[Click here to access/download;Figure;Fig.8 HSP laser output signal2.jpg](#)

Figure9

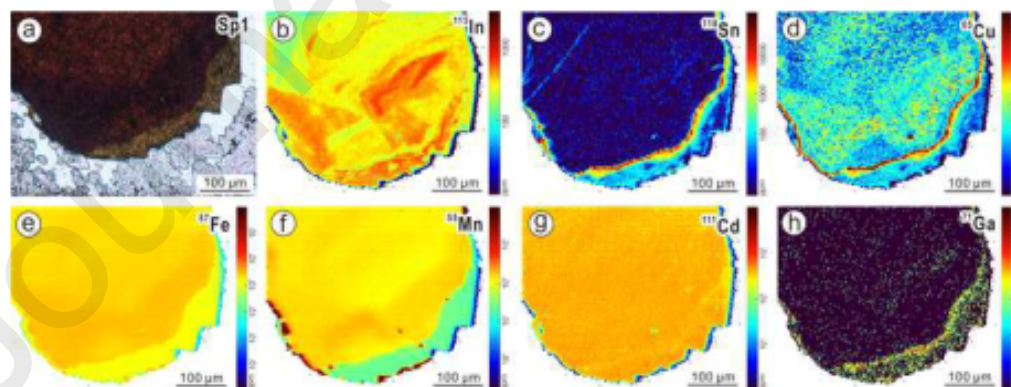
[Click here to access/download;Figure;Fig.9 HSP Laser mapping.jpg](#)

Figure10

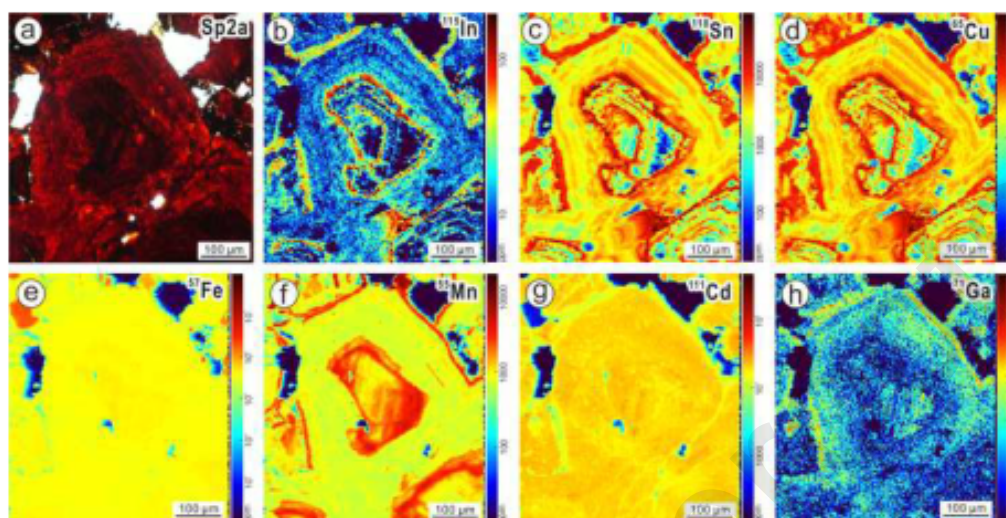
[Click here to access/download/Figure/Fig.10 HSP Laser mapping.jpg](#)

Figure 11

[Click here to access/download;Figure/Fig.11 HSP Binary plots.jpg](#)

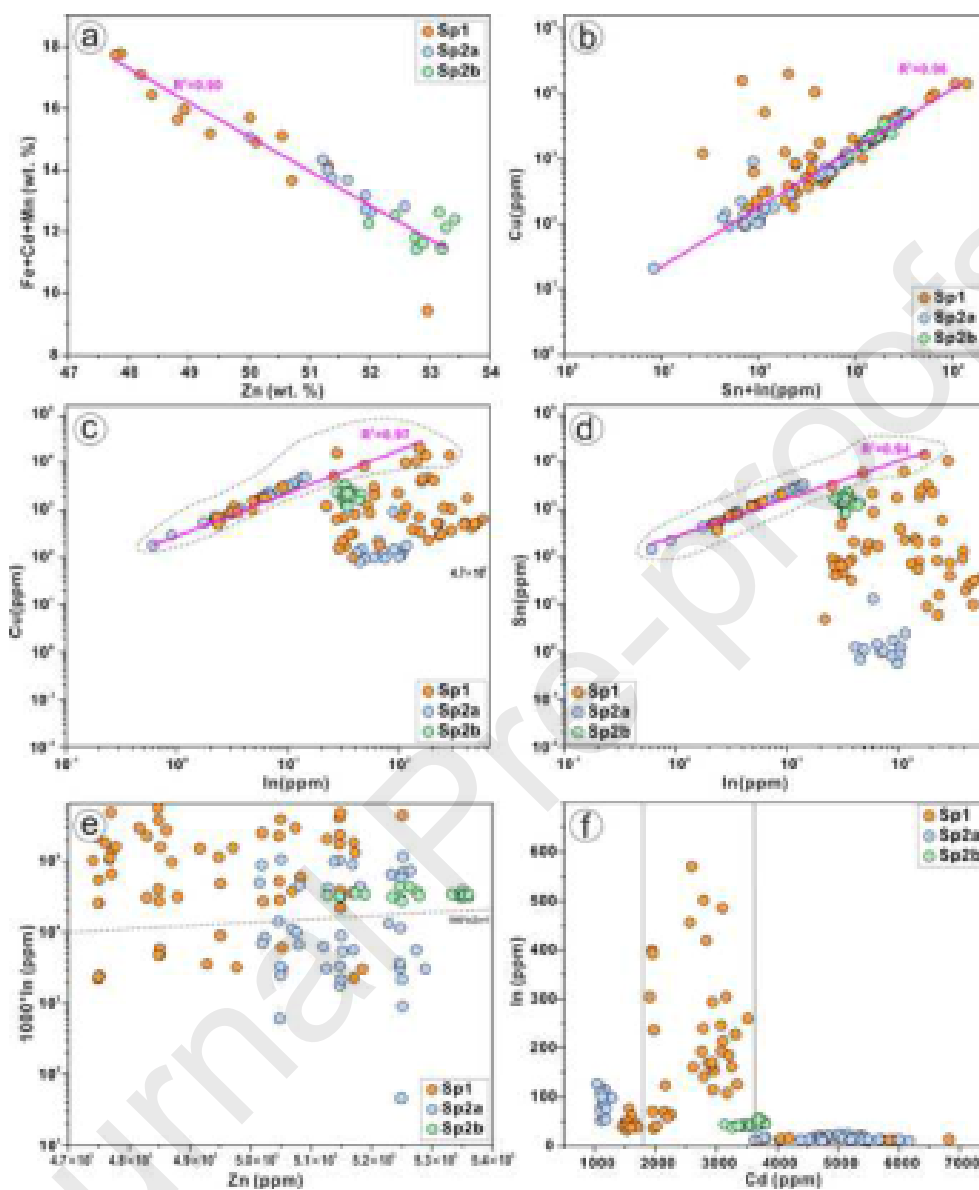


Figure12

[Click here to access/download;Figure/Fig.12 HSP temperature, sulfur fugacity, and thermodynamic.jpg](#)

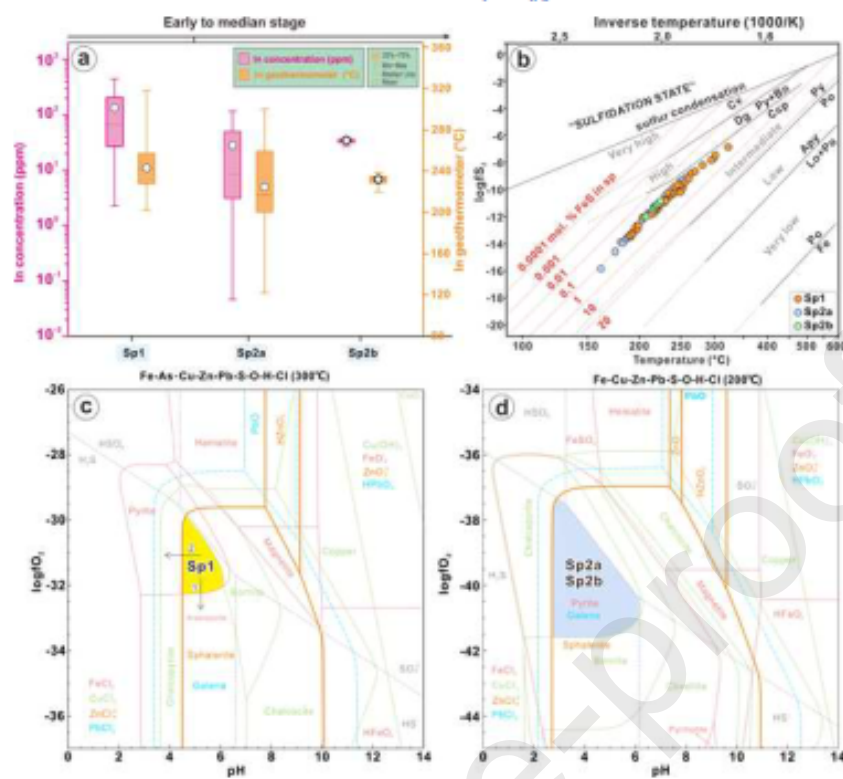


Figure13

[Click here to access/download;Figure/Fig.13 HSP Metallogenic model.jpg](#)

